BTS–IV(R/S)–06–22–0813 Reg. No. | | | | | | | | |

C

B.Tech. Degree IV Semester Regular/Supplementary Examination June 2022

ME 19-205-0404 APPLIED THERMODYNAMICS

(2019 *Scheme*)

Time: 3 Hours Maximum Marks: 60

Course Outcome

On successful completion of the course, the students will be able to:

CO1: Learn first and second law of thermodynamics and the application of these laws for various thermodynamic processes.

CO2: Apply the knowledge of thermodynamic laws to various engineering systems.

CO3: Analyse the performance of steam generators with additional accessories used for enhancing the performance.

CO4: Understand the various types of nozzles and its performance at various back pressure conditions and also for different flow conditions.

CO5: Calculate the mole and mass fractions of various mixtures and also various specific properties of the mixtures.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 –Analyze, L5 – Evaluate, L6 – Create

PO – Programme Outcome

(Use of Steam Table is permitted)

(Data not given may be suitably assumed. All such assumptions must be clearly stated)

PART A

(Answer *ALL* questions)

		$(8 \times 3 = 24)$	Marks	BL	CO	PO
I.	(a)	Explain the difference between high-grade energy and low-grade energy with suitable examples.	3	L2	1	1
	(b)	Explain the significance of Clausius inequality in thermodynamics.	3	L1	2	1
	(c)	Explain the difference between triple point and critical point.	3	L2	2	1
	(d)	Differentiate between fire tube boilers and water tube boilers.	3	L1	3	1
	(e)	Explain the effect of variation of back pressure in a steam nozzle.	3	L2	4	1
	(f)	Write a short note on velocity compounding of an impulse turbine.	3	L1	4	1
	(g)	Explain Dalton's law of partial pressure, Amagat's law of partial volumes and Gibbs-Dalton law for enthalpy, entropy and internal energy.	3	L2	5	1
	(h)	Explain Gravimetric and volumetric analysis and make a relation between the two analyses.	3	L2	5	1

PART B

		$(4 \times 12 = 48)$				
II.	(a)	Establish equivalence of Kelvin-Plank and Clausius statements.	6	L2	1	2,3
	(b)	A mass of 1.5 kg of air is compressed in a quasi-static process from 0.1 MPa to 0.7 MPa for which pv = constant. The initial density of air is 1.16 kg/m^3. Find the work done by the piston to compress the air.	6	L3	1	1,2,4

OR

(P.T.O.)

III.	(a)	Derive Maxwell equations and Tds equations from fundamentals	6	L2	2	1
	(b)	A heat pump working on the Carnot cycle takes in heat from a reservoir at 5°C and delivers heat to a reservoir at 60°C. The heat pump is driven by a reversible heat engine which takes in heat from a reservoir at 840°C and rejects heat to a reservoir at 60°C. The reversible heat engine also drives a machine that absorbs 30 kW. If the heat pump extracts 17 kJ/s from the 5°C reservoir, determine (i) The rate of heat supply from the 840°C source. (ii) The rate of heat rejection to the 60°C sink.	6	L5	1	1,2,4
IV.	(a)	Compare PV diagram of pure substances that contract during freezing with those substances which expands during freezing	6	L2	2	1
	(b)	In a reheat cycle, the initial steam pressure and the maximum temperature are 150 bar and 550°C respectively. If the condenser pressure is 0.1 bar and the moisture at the condenser inlet is 5%, and assuming ideal processes, determine (i) the reheat pressure (ii) the cycle efficiency (iii) the steam rate.	6	L3	3	1,2,4
		OR				
V.	(a)	Explain the working of binary vapor cycle with neat sketch.	6	L2	3	1
	(b)	Steam initially at 1.5 MPa, 300°C expands reversibly and adiabatically in a steam turbine to 40°C. Determine the ideal work output of turbine per kg of steam.	6	L3	2	1,2
VI.	(a)	Dry saturated steam at a pressure of 12 bar flows through the nozzles at the rate of 4.5 kg/s and discharges at a pressure of 1 bar. The loss due to friction occurs only in the diverging portion of the nozzle and its magnitude is 10% of the total isentropic enthalpy drop. Determine the cross sectional area at the throat and exit of the nozzle.	6	L4	4	1,2
	(b)	Explain Wilson line in connection with super saturated flow in nozzles.	6	L2	4	1,2
		OR				
VII.	(a)	Determine the mass flow rate of steam through a nozzle having isentropic flow through it. Steam enters nozzle at 10 bar, 500°C and leaves at 6 bar. Cross-section area at exit of nozzle is 20 cm^2. Velocity of steam entering nozzle may be considered negligible. Show the process on h-s diagram.	6	L4	4	1,2,4
	(b)	Analyse the effect of friction in nozzle flows.	6	L2	4	1,2
VIII.	(a)	Derive an expression for the mean value of gas constant for a mixture of gases	6	L2	5	1,2
	(b)	A mixture of 2 kg of methane and 4 kg of air contained in a vessel is at 1 bar and 15°C. The volumetric analysis of the air can be taken as 79% N_2 and 21 % O_2. Calculate for the mixture: (i) mass of methane, oxygen and total mass. (ii) mean value of gas constant and apparent molecular mass.	6	L4	5	1,2,4
		OR				
IX.	(a)	Derive an expression for the apparent molecular mass for a mixture of gases.	6	L2	5	1,2
	(b)	2 kg of oxygen at 2 bar and 40°C is mixed with 4 kg of nitrogen at 1 bar and 140°C to form a mixture. The process occurs adiabatically in a steady flow apparatus. Calculate the final temperature of the mixture. For nitrogen and oxygen, specific heat Cp may be taken as 1.04 kJ/kg K and 0.95 kJ/kg K respectively.	6	L3	5	1,2,4

Bloom's Taxonomy Levels
L1 – 7.5%, L2 – 52.5%, L3 – 25%, L4 – 10%, L5 – 5%

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